

# Deleting a Memory and Running It Again

*An instrument for checking what a model says about its own context*

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## Abstract

Ask a language model which part of its context made it answer the way it did, and it will tell you. Checking the answer means running the counterfactual — rebuilding the request without that item and asking again. Any API allows it; most stacks make it awkward enough that it is not done, because the context is flattened into a string and its items stop being addressable once the request is sent.

I built a runtime in which a session's context is addressable state, so a session can be forked, one named item removed, and the copy asked the same question again. That gives ground truth for the ablation: not what a model says its answer depended on, but whether the answer actually changes without the item. **This paper is about the apparatus** — what it takes to make a measurement like that honest, and what it found when I pointed it at six models from six labs, twice.

The registered hypotheses failed. What came out instead was sharper, and was then collected a second time on a corrected instrument, against four thresholds registered and timestamped beforehand. All four held. Every question and answer from both collections is saved, and every number recomputes from that record.

Models are reasonably good at saying *what* their answer rests on: 72–96% correct in the first collection and 72–85% in the second, above their own majority baseline in four of six models and then five of six. Their false positives are not random. Across both collections, 0 of 135 planted irrelevant notes were ever claimed to matter, 1 of 199 notes without figures was, and 62 of the other 63 fell on notes belonging to the question's arithmetic that did not decide it; the one exception is task-central and carries no quantity at all. **No own-context advantage was detected**: reasoning about your own context was no more accurate than reading a transcript of someone else's. And on a five-rung repair ladder, *telling* a model one of its notes is false helped every model that could be measured, while *then asking it to put the context right* moved one measurement of seven past the registered ceiling and left the others between −20 and +13 points.

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## 1. Introduction

This did not start as research. I was building a runtime for language-model agents in which the context — the pile of messages, notes and tool results that goes into each request — is proper state rather than a string that gets concatenated. Every item has an identity and a lifecycle. You can snapshot a session, change one thing, and resume it.

Once that existed, an experiment became possible that is otherwise very hard to run. If you want to know whether a model's answer really depended on a particular note, you can take the note out and ask again. Two copies of the same session, identical except for one item, both asked the same question. If the answers differ, the item was load-bearing. If they do not, it was not.

That is worth pausing on, though not for the reason I first wrote down. In fifty years of work on human self-report — starting with Nisbett and Wilson's *Telling More Than We Can Know* (1977) — the standing problem is that you cannot rerun a person with one memory deleted, so a stated reason can be shown *implausible* and never *wrong*. A

language model can be rerun, and prompt-level ablation and replay are not new; §8 names the closest published work, which makes the same move first. What the runtime adds is that the item is addressable rather than a substring, so the ablation is a named operation on state — cheap to run in bulk, and recorded item by item so somebody else can check it afterwards. That is a difference in convenience and auditability, not in what is possible.

So I used it to check a simple thing: when a model explains what its answer depends on, is it right?

I expected it not to be. I was wrong, and the ways in which I was wrong are the useful part.

But the measurements are not what I think is worth your time. Six models will be deprecated inside a year and their numbers with them. What survives is the apparatus and the discipline it needs, because **running the counterfactual is the easy half**. The hard half is everything that stops the result from being an artefact: what to baseline against, what to refuse to score, how to tell your own failure from the subject's, and how to know whether two runs were asked the same question at all. Every one of those is a decision I got wrong at least once, and §5 is the list.

The strongest evidence I can offer that the apparatus works is that it caught me, twice, in public, with dates. A result I had called the study's cleanest turned out to be an artefact of asking for a number the subject could not see — caught by a baseline the harness printed and I had not read. And a claim that deleting a false note was worth nothing weakened when one model gained a great deal from it. Both are in §4 and §6, because an instrument that cannot embarrass the person holding it is not measuring anything.

What this is not is a peer-reviewed study. I am a Rust engineer rather than a researcher, nobody who reads this literature regularly has checked it, and it should be read as a report on an instrument and what it measured. §7 says who did what, including the parts a language model did.

## 2. Method

### 2.1 The setup

Each trial plants a small dossier of notes in a fresh session — a set of records about, say, three warehouses — and asks a question the notes answer, such as which one runs out of space first. The model answers.

Then, for each note, two things happen:

- **The model is asked:** "if this note were removed, would you answer differently?"
- **The harness finds out:** it forks the session, removes that note, and asks the question again. Several copies, to check they agree with each other.

The copies never see the exchange in which the model already answered, so they are not being asked to remember, only to work it out again. The model's claim is then scored against what the copies actually did. Neither side gets special treatment: both are the same model, on the same context, minus one item.

### 2.2 The materials

Six dossiers, each with nine notes and a three-way question. Every dossier contains:

- notes that carry the arithmetic of the question (capacities, rates, stock levels),
- one note that decides the answer, usually a correction buried in prose,
- inert notes with no bearing on the question at all,

- and **two "red herrings"** — notes stating a plausible figure for each of the three options on a dimension that has nothing to do with the question. Road distances to a regional office. Rateable values. Deed reference numbers.

The red herrings exist because of a defect I found in my own material partway through. Before they were added, *every single inert note in the whole set had three digits or fewer*. So "contains numbers" and "matters" were tangled together, and a model that simply guessed "notes with figures in them are important" would have scored well for the wrong reason. With one red herring per dossier that shortcut still scored 0.83 against the truth — better than the models themselves managed — so I added a second. Two brings it to 0.74, and a test in the repository holds it there.

### 2.3 What is scored

The main measurement is deliberately narrow. It looks **only at notes whose removal did not change the answer** — where the treated copies and the control copies came out the same. For all of those, the correct claim is "no, this doesn't matter". So among items that all do nothing, does the model claim the numeric ones matter more often than the plain ones? What "came out the same" means exactly, and what a stricter reading of it costs, is §2.4.

Restricting it that way is the whole design. Across the material as a whole, notes with figures really *are* more likely to matter, because that is where the arithmetic lives. An unrestricted comparison would reward a model for a lucky prior. Inside a group of items that are all scored as inert, there is nothing left to be right about, so any gap between the halves cannot be knowledge.

What is scored, throughout, is a prediction of the **ablation outcome**: whether removing the item changes what the model says. That is not the same as whether the item influenced the original computation. An item can be read, reasoned over and still be redundant, and its removal will change nothing — the apparatus will call it inert, correctly, for the question it is asking. Every claim in this paper is about that question and no larger one.

### 2.4 What the apparatus refuses to do

Three refusals and one warning do more work than anything in the scoring, and each exists because something went wrong without it.

**It says, above the numbers, when the material did nothing to that subject.** Before the scores are printed, the harness checks that removing the decisive item actually moves the answer, that the control copies agree with each other, that the subject answered the dossier as its notes support, and — in the repair ladder — that the planted falsehood actually fooled it. These checks fire often, and every one is printed beside the numbers as an **unmet**: line rather than hidden in a footnote. In the repair ladder every rung is scored over all fifteen cells, the ones the falsehood never took included, which is why **carrying** reads as the share of cells the lie failed to fool and why the ladder is read as differences between rungs rather than as accuracies.

**One of those checks was registered as a gate, and the instrument never applied it.** §8.2 of the preregistration reads:

*If the control copies disagree with each other, the condition is reported with its instability figure and excluded from the primary analysis.*

The instrument reports the instability figure and excludes nothing. It compares the *unique plurality* of the readable copies on each side. A control that tied has no answer and drops out by construction; one that went two-one, or one readable answer against two that could not be read, still yields a plurality, and the condition stays in. The

published numbers in §4.3 are therefore the plurality-scored ones rather than the ones §8.2 specifies. They are kept as published, because that is what was run.

What it cost is small, and `cargo run --release` prints all three readings:

| reading             | items dropped | numeric        | plain       | herrings   | off-pivot    | discrimination |
|---------------------|---------------|----------------|-------------|------------|--------------|----------------|
| as published        | —             | 32/140, 31/136 | 0/101, 1/98 | 0/69, 0/66 | 32/32, 31/31 | 6/6, 6/6       |
| §8.2 as registered  | 35, 33        | 26/119, 25/117 | 0/87, 0/84  | 0/59, 0/58 | 26/26, 25/25 | 6/6, 6/6       |
| both arms unanimous | 57, 55        | 21/103, 19/99  | 0/81, 0/80  | 0/51, 0/52 | 21/21, 19/19 | 5/6, 6/6       |

Applying the registered gate drops 35 items in the first collection and 33 in the second, and moves no conclusion. All four replication thresholds hold and the direction is unanimous both times. It does remove the one plain-note false positive, which sat on a material whose control had not agreed: under §8.2 the plain count is 0 of 171 rather than 1 of 199.

The third row asks the same of the treated batch. Nothing registered that, so it is a sensitivity analysis chosen after the fact and labelled as one. It is also the only row where something gives, and what gives is the unanimity of direction in the first collection, 6 of 6 becoming 5 of 6.

**It refuses to let a copy see the answer already given.** A session that has committed to an answer and is then copied would be reading its own commitment; both arms are blinded to the exchange in which the subject already answered, so what differs between them is the item and nothing else.

**It refuses to conflate its own failure with the subject's.** A turn that ran out of output tokens having said nothing is not a subject declining to commit — it is a question that never got an answer, and scoring it as a wrong claim would charge a model for my token ceiling. Those are counted separately and excluded. An unreadable answer from a subject that *did* speak is the subject's failure and is scored as one.

**It refuses to let two runs be compared unless they were asked the same thing.** Every experiment fingerprints the exact text of its questions and its material. When the location probe came out, two of the seven fingerprints moved and five did not — so a v4 repair figure and a v5 one are provably the same question, and a v4 attribution figure and a v5 one provably are not.

### 3. What was fixed in advance

The hypotheses, the numeric predictions, the exclusion rules and a nine-row table of "if the results look like *this*, do *that*" were written down before collection and committed to a separate public branch.

This matters more than usual here, because I generated the hypotheses by looking at some early exploratory runs. That is a legitimate way to *find* a hypothesis and a terrible way to *test* one, so the fix is to register it and then collect fresh data. Which is what happened.

Being precise about the timestamp, since a preregistration is worth exactly what its date is worth. The registered document records its decision table as committed at 18:54:00Z on 2026-09-03 — 76 seconds after the first model's `attribution` report was written, and 78 and 88 minutes before that model's `repair` and `instrumented` reports. That commit is not in the published history: the instrument was split out into its own crate the following morning and its history remade with it, so the hash the document cites resolves to nothing and those three margins rest on my word.

What anyone can check is the first commit on the public `preregistration` branch, at 20:38:11Z — sixteen minutes *after* the last of the first model's reports, and eight minutes *before* the first request went out for the other five. So for one model of six the public record puts the registration after the data, and for the other five before it, by a margin nobody would call comfortable.

Two things cut against even that. A commit's timestamp is set by the machine that made it rather than by the server, and GitHub does not publish when a push arrived, so "public" here means anyone can see the same timestamps I can, not that a third party vouches for them. And "I hadn't looked yet" is not something anyone can verify either; it is not offered as evidence. For the first collection the honest claim is the weaker one, and the repository says so too.

**The second collection repairs that.** After the defect in §4.2 was found, an amendment naming four thresholds — what would count as a replication, decided before there was anything to look at — was committed to that public branch at 10:18:58Z, and the first request of that collection went out at 10:31:43Z. Thirteen minutes, both timestamps visible to anyone. The endpoint whose first-collection margin cannot be checked has a checkable one the second, which is the entire reason the re-run was worth \$15.

The registered document also contains a rule I want to draw attention to, because it is the rule that stopped me rescuing my own hypothesis:

┆ *No new hypotheses. If all of them fail, the honest paper is a negative one.*

All but one of them failed, and no replacements were invented — §4.6 is the test of that rule rather than an exception to it, since the most interesting thing measured here carries no prediction and is fenced off as a pilot. What the second collection then established is narrower than a hypothesis and more useful than a null: the four registered thresholds held, so the effect the first collection found is not an artefact of the defect it was found with.

## 4. Results

Six models: `upstage/solar-pro4`, `deepseek/deepseek-v4-flash-0731`, `tencent/hy3`, `meituan/longcat-2.0`, `x-ai/grok-4.6`, `z-ai/glm-5.3-flash`. Six labs, and a range from a model that emits an answer with no visible deliberation to a frontier one. Each was measured twice: a first collection on instrument v4, run on 2026-09-03 and into the following morning, and a second on v5 on 2026-09-04, after the defect in §4.2 was found and removed.

**The confirmatory claims are the four thresholds in §4.3, registered and timestamped before the second collection.** Everything else here is exploratory or harness validation, is labelled as such where it appears, and is not used to extend those claims.

**How to read the tables.** A *dossier* — the harness calls it a *material* — is one planted context: nine *notes*, a three-way question, and one answer the notes support. An *item* is one note in one session, and it is what an ablation is run on. A *batch* is the set of copies asked under one condition, control or ablated. A *cell* is one dossier in one run of the repair ladder, and a *rung* is one stage of that ladder. An *arm* is one side of the own-versus-foreign comparison in §4.4. A *collection* is one sweep of all six models: the first on instrument v4, the second on v5. Throughout, denominators count the items actually measured, which is why they differ between models where a turn was cut, and numerators count claims that an item mattered.

4.1 Models are decent at saying what mattered

The baseline is the best a subject with no self-knowledge could score by picking one answer and repeating it — not chance. Both collections are exploratory here; no prediction covers accuracy.

| model             | first collection | baseline | second collection | baseline |
|-------------------|------------------|----------|-------------------|----------|
| glm-5.3-flash     | 52/54 — 96%      | 76%      | 46/54 — 85%       | 70%      |
| hy3               | 48/54 — 89%      | 76%      | 46/54 — 85%       | 74%      |
| grok-4.6          | 47/54 — 87%      | 69%      | 44/54 — 81%       | 69%      |
| longcat-2.0       | 44/54 — 81%      | 81%      | 46/54 — 85%       | 81%      |
| deepseek-v4-flash | 41/52 — 79%      | 67%      | 39/53 — 74%       | 66%      |
| solar-pro4        | 39/54 — 72%      | 80%      | 39/54 — 72%       | 74%      |

Four of six beat their baseline the first time and five of six the second — `longcat-2.0` moves from level with it to above it, and `solar-pro4` is the one model below it in both. This is the first place my expectations broke: any framing built on "models cannot report what their answers depend on" is contradicted by my own data.

4.2 An item's number was not a fair question

The same models were asked, about the same notes: *what number is this note in your context?*

| model             | correct |
|-------------------|---------|
| solar-pro4        | 2/18    |
| deepseek-v4-flash | 0/18    |
| hy3               | 0/18    |
| longcat-2.0       | 0/18    |
| grok-4.6          | 0/18    |
| glm-5.3-flash     | 0/18    |

2 of 108. An earlier draft called that the study's cleanest result. It is not a result at all.

The projector renders each item as `label:` and its content, never a number, and `attribution` installs no handles — so these subjects had no way to see the numbering they were being asked about. They were asked for a value they could not observe. The score says so: 1.9% against a 33% majority baseline, an order of magnitude *below* what a constant guesser gets, which is the signature of an unanswerable question rather than an absent faculty (§5, point 1).

nachalnik *can* address items to a model — `inspect`'s `look` lists every one by number — and that handle simply was not granted here. **So this says nothing about whether a model can locate an item when allowed to look.** That question is open; the figure was withdrawn on 2026-09-04, and v5 removed the probe rather than guess at it.

### 4.3 The errors have a shape

Looking only at notes classified as inert under the scoring rule of §2.4 — control and ablated batches landing on the same unique plurality answer:

**First collection, instrument v4 — exploratory.** The hypotheses were registered, but the defect in §4.2 was still in the instrument when these were taken.

| model             | numeric claimed | plain claimed | red herrings | other numeric |
|-------------------|-----------------|---------------|--------------|---------------|
| longcat-2.0       | 10/26           | 0/18          | 0/12         | 10/14         |
| solar-pro4        | 9/27            | 0/16          | 0/12         | 9/15          |
| deepseek-v4-flash | 4/19            | 0/16          | 0/10         | 4/9           |
| hy3               | 4/24            | 0/17          | 0/12         | 4/12          |
| grok-4.6          | 3/21            | 0/16          | 0/11         | 3/10          |
| glm-5.3-flash     | 2/23            | 0/18          | 0/12         | 2/11          |
| <b>total</b>      | 32/140          | 0/101         | 0/69         | 32/71         |

"Other numeric" is the residual: every inert note carrying two adjacent digits that is not on its dossier's planted-decoy list. It is the question's own arithmetic in all but one case, named below; the denominators count every such note measured, over-claimed or not.

**Second collection, instrument v5 — confirmatory.** Same six models, same material, location probe removed. The four thresholds these are read against were committed to a public branch thirteen minutes before the first request.

| model             | numeric claimed | plain claimed | red herrings | other numeric |
|-------------------|-----------------|---------------|--------------|---------------|
| longcat-2.0       | 8/26            | 0/18          | 0/12         | 8/14          |
| solar-pro4        | 7/24            | 0/16          | 0/12         | 7/12          |
| deepseek-v4-flash | 4/19            | <b>1/16</b>   | 0/10         | 4/9           |
| hy3               | 5/23            | 0/17          | 0/11         | 5/12          |
| grok-4.6          | 4/21            | 0/16          | 0/10         | 4/11          |
| glm-5.3-flash     | 3/23            | 0/15          | 0/11         | 3/12          |
| <b>total</b>      | 31/136          | 1/98          | 0/66         | 31/70         |

Three counts here, and a caveat on the third. None of them is a rate, and all three span both collections.

**Not one model, in either collection, claimed a red herring mattered.** 0 of 135. These six models are not fooled by irrelevant numbers.

**One note without figures was claimed, out of 199.** The first collection had none at all; `deepseek-v4-flash` claimed one in the second, which is why this sentence no longer says "never".

**62 of the 63 false positives carrying a figure were notes belonging to the question's own arithmetic** that happened not to decide it. Every model showed it in both collections, and the effect ran the same way for all six

each time (sign test; Appendix A gives  $p = 0.031$  two-sided for the first collection and  $p = 0.016$  one-sided for the second, where the direction was registered in advance).

**The remaining one is worth naming, because the category is looser than its name.** The instrument sorts a numeric inert note into "red herring" if it is on the dossier's planted-decoy list and into "the question's arithmetic" otherwise, and "numeric" means any two adjacent digits anywhere in the text. `mill/records/guidance` — "*Assessors read the Luza column first: Luza has been the binding constraint in every quarter since 2019*" — trips that test on the year and carries no quantity at all. `longcat-2.0` claimed it once, in the first collection. The other 62 sit on a capacity, an intake, a rate, a stock or a threshold the question really is computed from, so the finding survives. But the accurate name for the category is *notes central to the task that did not decide it*: "arithmetic" describes 62 of the 63 rather than defining the bucket. Counting the plain note above, that is 64 endpoint false positives in total.

My registered hypothesis was that models over-weight *digits*. That is wrong. They over-weight *things that look like the reasoning*, which is a different and much more sensible failure. I registered that distinction in advance as the outcome most likely to be mistaken for the other one, which is the only reason I am able to say it cleanly now.

Where the errors land is a count and stands. *Why* they land there is a reading, and a note that belongs to the question's arithmetic differs from an inert one in more than topic — length and position travel with it, and this material was not built to pull those apart.

#### 4.4 Reading your own context looks like just reading

**No own-context advantage was detected.** Each model was asked the same kind of counterfactual question twice: once about its own context, and once about a transcript of a different session's context, quoted to it. Five items per arm, less any turn that was cut — which is why `deepseek-v4-flash` has a denominator of 4 in the first collection.

| model             | own, first | foreign, first | own, second | foreign, second |
|-------------------|------------|----------------|-------------|-----------------|
| solar-pro4        | 1/5        | 3/5            | 1/5         | 3/5             |
| deepseek-v4-flash | 1/5        | 1/4            | 3/5         | 3/5             |
| hy3               | 3/5        | 3/5            | 4/5         | 3/5             |
| longcat-2.0       | 3/5        | 3/5            | 3/5         | 5/5             |
| grok-4.6          | 3/5        | 2/5            | 3/5         | 3/5             |
| glm-5.3-flash     | 4/5        | 4/5            | 5/5         | 5/5             |
| pooled            | 15/30      | 16/29          | 19/30       | 22/30           |

Reflecting on its own context bought a model nothing: three exact ties each time, and pooled it came out slightly worse both times. Nor are the two arms interchangeable per model. The own-minus-foreign difference falls within the registered ten points on four of six models the first time and three of six the second, and where it misses it misses by twenty to forty points, in both directions.

What holds without exception is the weaker thing: every per-model interval contains zero, both times, as does each pooled one. At five items per arm per model this is a failure to detect. It is not the sign of an effect, not equivalence between the arms, and not evidence that privileged access is impossible.

## 4.5 Telling beats measuring

The last experiment plants a false note that contradicts the records, then asks the same question five times, adding exactly one thing between each pair:

1. **carrying** — it just has the false note
2. **again** — the same question, a second time, nothing added (a control)
3. **unprompted** — it is given tools to inspect and edit its own context, with no hint anything is wrong
4. **told-so** — it is told one of its notes contradicts the records, and names which
5. **repaired** — it is asked to fix it

Five of the six dossiers (the ladder does not use `mill`) with three independent ladders each: fifteen observations per rung per model, less the odd cell a truncated turn took out. The contrasts quoted after the tables are paired over the cells where both rungs were measured, as registered, so wherever a turn was cut they are not differences of these columns. **First collection:**

| model             | carrying | again | unprompted | told-so | repaired | used the tools |
|-------------------|----------|-------|------------|---------|----------|----------------|
| solar-pro4        | 5/15     | 5/15  | 0/15       | 1/15    | 2/15     | 27%            |
| longcat-2.0       | 3/15     | 3/15  | 2/15       | 7/15    | 2/15     | 37%            |
| deepseek-v4-flash | 2/15     | 1/15  | 5/15       | 8/15    | 10/15    | 50%            |
| hy3               | 3/15     | 6/15  | 6/14       | 14/15   | 11/15    | 62%            |
| grok-4.6          | 4/15     | 3/15  | 4/15       | 9/15    | 9/15     | 67%            |
| glm-5.3-flash     | 8/15     | 6/15  | 9/15       | 10/15   | 10/14    | 60%            |

### Second collection:

| model             | carrying | again | unprompted | told-so | repaired | used the tools |
|-------------------|----------|-------|------------|---------|----------|----------------|
| solar-pro4        | 6/15     | 6/15  | 0/15       | 3/15    | 0/15     | 25%            |
| longcat-2.0       | 3/15     | 2/15  | 2/15       | 3/14    | 3/15     | 37%            |
| deepseek-v4-flash | 3/15     | 1/15  | 1/15       | 7/14    | 8/14     | 47%            |
| hy3               | 6/15     | 7/15  | 11/15      | 13/15   | 13/15    | 62%            |
| grok-4.6          | 5/15     | 5/15  | 5/15       | 9/15    | 9/15     | 68%            |
| glm-5.3-flash     | 9/15     | 7/15  | 8/15       | 10/15   | 14/15    | 58%            |

Models using the tools on fewer than half the questions they could have were excluded from this comparison in advance: two of six the first time, and three the second, because `deepseek-v4-flash` came in at 47% having cleared the same bar at exactly 50%. The rule is pre-specified and was applied as written rather than nudged — see §5, point 10. `solar-pro4` is worth a sentence on its own: in the first collection it made 16 edits off 6 tests, rewriting its own context nearly three times as often as it measured anything, and 13 off 7 in the second; both times its answers got worse with tools than without them (5/15 down to 0/15, then 6/15 down to 0/15). That is a small, concrete warning about giving edit authority to weak models.

Among those that cleared the bar — four models the first time, three the second — here are the three registered contrasts in percentage points, paired over the cells where both rungs were measured:

| model                    | tools, no hint | told | asked to fix |
|--------------------------|----------------|------|--------------|
| <i>first collection</i>  |                |      |              |
| deepseek-v4-flash        | +27            | +20  | +13          |
| glm-5.3-flash            | +20            | +7   | +7           |
| grok-4.6                 | +7             | +33  | 0            |
| hy3                      | +7             | +50  | -20          |
| <i>second collection</i> |                |      |              |
| glm-5.3-flash            | +7             | +13  | +27          |
| hy3                      | +27            | +13  | 0            |
| grok-4.6                 | 0              | +27  | 0            |

**Being told helped every model, both times.** Four out of four, then three out of three.

**Being handed tools with no hint helped more than registered.** P3 put the ceiling at five points. Every model in the first collection cleared it, as did two of three in the second. Three of those six gains are seven points, and two of the three do not survive the reading below.

**Being asked to put the context right moved one measurement of seven.** glm-5.3-flash went 10/15 to 14/15 in the second collection; one model went twenty points the other way; the rest sat between. At  $n = 15$  none of these is a precise estimate, and the registered one-sided ceiling is not an equivalence test. What the ladder supports is that the instruction did not reliably help, not that its effect is zero.

**Restricting the reading to the cells the lie actually fooled changes no direction.** The tables above score every cell, including the ones where the falsehood never took (§2.4). The lie took hold in 43 of 60 cells in the first collection and 25 of 45 in the second, among the models that cleared the bar. Over those cells the dilution comes off and the pooled contrasts sharpen:

| contrast       | first collection        | second collection      |
|----------------|-------------------------|------------------------|
| tools, no hint | +14                     | +16                    |
| told           | +37 (17 gained, 1 lost) | +32 (8 gained, 0 lost) |
| asked to fix   | -2                      | +12                    |

The two small seven-point gains from the tools disappear under this reading — grok-4.6's in the first collection and glm-5.3-flash's in the second — because the single cell each came from was a cell the lie had never fooled. cargo run --release prints both readings.

**The registered prediction was that the deletion would be what mattered, and this ladder cannot say whether it does.** It never withheld deletion. The tools arrive at unprompted and are never taken away, so told-so and repaired differ in what the subject is told, not in what it is allowed to do: the last rung adds the instruction "Something in your context is wrong. Find out what, and put it right." That is deliberate, because a rung that also

handed over a new capability would confound the two. What the ladder can say is that *disclosure* moved the answer and the *instruction to act on it* mostly did not — for the answer the subject is about to give, which is the only thing it scores (§5.1). No sentence here should be read as measuring the value of granting edit authority.

The subjects bear it out. Across both collections they made 10 edits over the two rungs where they held the handles unasked: 4 at `unprompted`, and 6 after naming the note at `told-so`. They made 228 once they were asked to put the context right. Naming moved a few of them to act; being told to act moved them roughly twenty times as often. The rungs always run in this order and the later ones have had more turns in which to edit, so that ratio describes the ladder as run rather than a controlled comparison.

## 4.6 A capability, measured and not predicted

**No prediction covers anything in this section.** It was registered as harness validation — a check that the ablation handle returns what it should — and that is the only claim it supports. It is reported because leaving it out would be worse, and it is fenced because the last figure this paper reported without a prediction attached turned out to be an artefact (§4.2).

`instrumented` asks the same counterfactual three ways: of a subject with no handle, of that same subject once given one *after* it has already answered, and of a fresh subject given one that was never asked to guess.

| model                      | reported | retested | tested | reported → retested | reported → tested      |
|----------------------------|----------|----------|--------|---------------------|------------------------|
| <code>glm-5.3-flash</code> | 45/53    | 44/53    | 48/54  | -2 (p = 0.73)       | +4 (p = 0.38)          |
| <code>hy3</code>           | 43/54    | 45/54    | 49/54  | +4 (p = 0.36)       | +11 (p = 0.09)         |
| <code>grok-4.6</code>      | 36/54    | 39/54    | 47/54  | +6 (p = 0.29)       | <b>+20 (p = 0.013)</b> |

Every model used the handle on essentially every question — 98%, 100%, 100%, and 353 tests between them — so whatever this is, it is not a failure to reach for the tool.

Three of three point the same way on `reported` → `tested`. On `reported` → `retested` it is two up and one down, none of the three significant, and the first collection's single `glm-5.3-flash` run — the stage-1 harness check — had the same shape: level on `retested`, up on `tested`. But **tested differs from reported in two ways at once**: it has the instrument *and* it never committed to an answer first. Nothing here can say which of those is doing the work, and the obvious fourth arm — a subject that has not spoken and has no instrument — does not exist in this design.

So this is a pilot for an experiment, not an experiment. It is written down here with its confound named so that the registered version has something to be compared against.

## 5. What it takes to measure this honestly

Forking a session and removing an item is a weekend's work. Everything below is what the rest of the time went on, across two collections, and each line is something I got wrong at least once.

**1. Baseline against the best constant answer, not against chance.** This is the one that caught me. The location probe scored 2 of 108 against a *majority* baseline of 33% — the score a subject with no self-knowledge gets by picking one answer and repeating it. Landing an order of magnitude under the do-nothing baseline is not a weak faculty, it is a broken question, and the harness had been printing that number all along.

**2. Test your preconditions and print them.** In the second collection the harness flagged 32 of the 90 repair-ladder cells because the planted falsehood never fooled the subject in the first place. Those cells stay in the tables — the ladder is read as differences between rungs — and the flag is what stops a model too sharp to be fooled from reading as a model that failed to repair. §4.5 gives the ladder both ways.

**3. Restrict the endpoint to items where the honest answer is known.** The primary measurement looks only at notes whose removal did not change the scored answer, because there the correct claim is "no" for every item. Across the material as a whole, notes with figures really are more likely to matter — an unrestricted comparison would pay a model for a lucky prior.

**4. Ask about two copies, not about the model.** "Would *your* answer change" is a different question from "will these two copies differ", and a subject can be exactly right about the copies and scored wrong because the live session — which has the elicitation in its context — answered unlike both. I learnt this from a model that was right while I was marking it wrong.

**5. Blind the copies to the answer already given.**

**6. Separate your failure from the subject's.** A truncated turn is your token ceiling, not a wrong claim.

**7. Fingerprint the questions.** Otherwise "we re-ran it" is an assertion. When the location probe came out, two of seven fingerprints moved and five did not, and the record says which.

**8. Grant looking and changing separately.** A tool that answers "may it experiment on itself" must not also answer "may it rewrite its own memory". One model in this cohort made 16 edits off 6 tests and got worse.

**9. Use one frozen origin for both measurements.** When a subject tests its own context, it should fork from the same snapshot the harness scores against — otherwise its evidence and your ground truth are two different experiments.

**10. Apply your gates as written, even when they cost you.** The handle-use gate excluded a third model in v5 because it came in at 47% against a pre-specified 50%, having been at exactly 50% the first time. Three points, one model, no adjustment.

**11. Correct for clustering.** Nine items from one dossier are not nine independent observations.

**12. Write down what would count as a replication before you collect it.** The only real degree of freedom in a re-run is what counts as the same answer, and without it fixed in advance, close numbers read as replication and divergent ones read as explanation. Both feel honest at the time.

## 5.1 What the measurements say

Secondary to the above, and shorter-lived, but they are what the apparatus was pointed at.

**Expect false positives in a specific place.** Across 135 planted irrelevant numbers over two collections, models fell for none. One note without figures was claimed, out of 199. Every other mistake landed on material that was topically central and causally inert. If you ask a model which parts of its context mattered, that is where its errors will be.

**Don't assume self-inspection is privileged.** No own-context advantage was detected: a model reasoning about its own context did no better than the same model reading a stranger's transcript — pooled, slightly worse. At five items per arm per model this is a failure to detect, not a demonstration that none exists.

**Tell your agents things.** There is enthusiasm for handing agents tools to prune their own context. On this evidence the *information* carries more than the instruction to act on it: saying that a note is false helped every model measurable, while then telling it to put the context right usually did not help — one measurement of seven cleared the registered ceiling, one went twenty points the other way, and the rest sat between. The tools were in its hands throughout both rungs, so this compares two things a subject was told, not a subject with handles against one without.

That last reading has a limit worth stating plainly, because I stated it too broadly first. **The ladder scores the answer a subject is about to give, not what it carries afterwards.** A ladder ends; a session does not. Removing a falsehood can be worth little to the next answer and a great deal to every answer after it, and nothing in this design can see the difference.

**If you ask a model to name an item, give it a way to look first.** Asked for an item's number with no handle to inspect its context, every model answered confidently and wrongly rather than declining. Whether they manage it *with* a handle is untested here, and is the obvious next experiment.

## 6. Limitations

I would rather write these than have them found.

**Six models, one API, two collections, one pair of hands.** The second collection is a replication in the sense that matters — fresh data against thresholds fixed beforehand — and in no other: same six models, same provider, same author, a day apart. Provider routing was not controlled. Nobody independent has run this.

**The materials are synthetic and all of one shape** — three options, an arithmetic question, a buried correction. Whether any of this generalises to real agent contexts, with tool output and long histories, is unknown and not claimed.

**The foreign arm arrives quoted where the subject's own arrives as context.** That is the difference §4.4 tests and also, unavoidably, a difference in presentation. The quotation is rendered exactly as the projector renders the subject's own notes, which is the mitigation; the registered counterbalance — running the two dossiers the other way round — was never run, in either collection. The null in §4.4 is read with that confound unremoved.

**The hypotheses came from exploratory data.** I found them by looking at early runs, then registered them and collected fresh data. Standard practice, but it means the exploratory runs cannot also count as evidence, and they do not.

**The first collection's registration is weaker than the second's.** The commit the registered document cites for its decision table is no longer in the published history, and the first commit on the public branch postdates one model's complete first-collection results and predates the other five by eight minutes (§3). The second collection's thresholds were on that branch thirteen minutes before its first request, so that endpoint has a margin anyone can check; the first collection's figures keep the weak one.

**Two models were substituted mid-study,** both logged with dates and reasons, and neither had produced a figure bearing on any hypothesis. One could not complete a run through its provider, which cut it off after about fifteen requests every time; one was swapped for a stronger and cheaper alternative in the same slot.

**One reported result was withdrawn after drafting, on 2026-09-04.** An earlier version of this paper treated the 2-of-108 location figure as its cleanest finding. It is an artefact of asking for a number the subject had no way to see; §4.2 now says so and the abstract no longer mentions it. It was caught while scoping a follow-up study, which

is later than it should have been. The class of error is fenced now rather than merely regretted: `tests/machinery.rs` requires that any experiment asking for an item's number also grant a handle to look at the numbering, and pins the two frozen experiments that do not.

**The withdrawn probe ran first — checked, and it did not matter.** The v5 collection removed it and re-ran the whole study: the pooled difference moved from +23 to +22, discrimination stayed negative on 6 of 6, and every registered threshold held (§4.3, `RESULTS-v5.md`). The concern below stood until it was tested and no longer does; it is kept because a reader should see what was suspected as well as what was found.

**The withdrawn probe ran first, and its answers stayed in the context.** `attribution` asks the three location questions *before* the counterfactual battery, so every claim in §4.1 and §4.3 was made in a context where the subject had just produced three confident, wrong item numbers. This is identical across all six models and both arms, so comparisons within the study are unaffected, and the §4.3 result is a contrast between two kinds of item in the same context, where any general shift in confidence cancels. The absolute rates could still be moved by it. I would not order the probes this way again.

**The ablation handle is now confirmed on three model families.** v5 registered this check on three models; the first attempt died on an exhausted key limit and it was re-run on 2026-09-04 once that was raised. Handle use came in at 98%, 100% and 100% of questions across `glm-5.3-flash`, `hy3` and `grok-4.6`, with 353 tests run and no edits. The limitation immediately below is therefore **closed**, and is kept in place so that a reader comparing versions can see what changed. `PREREGISTRATION.md` §11 has the dates.

**Only one model ran the tool-use check** that confirms the ablation handle returns what it should. It did — 100% use across 108 questions — but a second would have been better.

## 7. How this was made

Given the subject matter, this section is not optional.

**The great majority of this work was done by a language model** (Claude Opus 5), working interactively with me over one long session. It wrote the runtime harness, the experiment code, the statistics, the preregistration, the analysis tooling, and the first draft of this paper. I set the direction, made every judgement call about scope and spending, funded it, and pushed back — several times decisively, including on the framing of related work and on the choice of models.

I am a Rust engineer, not a researcher. This is an independent technical report rather than a peer-reviewed study, and its status should be read that way.

What makes it checkable regardless of who wrote it is that **you do not have to trust the process**:

- the preregistration is on a timestamped public branch, dated before most of the data existed;
- every question put to every model and every answer received is saved verbatim;
- every number in this paper recomputes from those records, and the primary endpoint, the four replication thresholds and both readings of the repair ladder come out of one command;
- the deviations log has twenty-one dated entries, including the ones that are unflattering.

Several real errors were caught during the work and are recorded rather than tidied away: a primary endpoint that turned out to be circular (the tool printed the answer it was being scored on), a material set that could not falsify its own hypothesis, an analysis script that silently dropped a model from the results, and a timestamp claim that

was 76 seconds wrong and now rests on a commit the published history no longer holds (§3). I mention these because a paper with no visible mistakes in a process this long is a paper whose mistakes were not written down.

## 8. Related work

**Apparatus.** Ground truth about what caused something requires intervention rather than observation (Pearl, 2009), and that intervention has been applied to language models at every layer, this one only lately. Causal mediation analysis (Vig et al., 2020), interchange interventions (Geiger et al., 2021) and activation patching (Meng et al., 2022) intervene on internal representations. At the prompt level, ERASER (DeYoung et al., 2020) scores a rationale by erasing the tokens it names and measuring what changes — *comprehensiveness* and *sufficiency* — which is this measurement performed on a static string. What a runtime adds is that the string is a live session: context is addressable state, so a session can be forked, a named item removed, and the copy resumed from the same frozen point with everything else held identical.

The nearest neighbour is Causal Agent Replay (Shah, 2026), which treats a run as a structural causal model, applies `do(·)` to a step and re-executes forward under the same policy — the same manoeuvre every claim here is scored against. Two differences matter: it intervenes on *steps* rather than on named context items, and it uses no self-report at all, validating against planted ground truth instead. The comparison this paper is built on, between what a model says and what a fork does, is not one it attempts.

VISTA (Xu et al., 2026) reaches the same diagnosis from the other end. Its first contribution is that frontier models are "proprioceptively blind to their own context", and its answer is a dashboard of per-block metadata with archive and recover tools. That is concurrent work, arrived at independently and found during the literature review rather than built on, and the variable it makes visible is not this one: VISTA's is *magnitude* — block size, recency, remaining budget — where this is *causal dependence*. The distinction is one of kind rather than degree. No amount of metadata display answers "would I have said something else without note 4", because a counterfactual has to be run. VISTA makes context legible; this makes it testable.

Treating context as addressable memory is otherwise well established, for other purposes. MemGPT (Packer et al., 2023) manages it as OS-style virtual memory with paging and eviction and LLMingua (Jiang et al., 2023) compresses it; Active Context Compression (Verma, 2026) has an agent consolidate a trajectory and delete the raw logs, scored on tokens saved at equal accuracy; Self-GC (Hao et al., 2026) folds, masks and prunes indexed context objects by predicted future usefulness rather than by provenance; Slipstream (Chen et al., 2026a) validates a compaction summary against the agent's independently continued reasoning. The premise is shared and the purpose is not: the same primitives are used here to verify a claim rather than to fit a budget.

**Findings.** Self-report about context is already benchmarked, and the accuracy of such reports is not a new question. Zeng et al. (2026) put nine kinds of question to models about their own behaviour — whether a prompt edit would change the answer is one of them — check the answers by resampling, find the skill real but limited, and then train models to be better at it. Naphade et al. (2026) run a third-party control and report a small significant self-advantage. This is not a new benchmark of that, and does not claim to be. One part of Zeng et al. is closer still. To make training data, their pipeline rewinds a conversation to an earlier turn, changes one thing, replays it, and has a judge score what the model then does. That is the move this apparatus makes, and it is published first. The purpose is the opposite: there the fork manufactures examples and a model is trained on them; here nothing is trained, and the fork is the measurement. What is asked differs too. Their model predicts a judge's score for an edit it is told about; here the subject is asked about the context it is sitting in, and the copies that check it have one item removed rather than rewritten.

That post-hoc explanations are frequently unfaithful to what actually drove a prediction is settled too (Jain and Wallace, 2019; Jacovi and Goldberg, 2020; Atanasova et al., 2023). Turpin et al. (2023) show models producing plausible rationales that never mention the feature actually steering them, and Lanham et al. (2023) show that perturbing chain-of-thought steps sometimes leaves the decision unmoved, with large variation by task. §4.3 sharpens the shape rather than the fact. The false attributions are not drawn to digits as such — not one model claimed a planted numeric red herring, 0 of 135 — but to notes belonging to the question's own arithmetic that happened not to decide it: 62 of the 63 numeric false positives, the exception being a note that is central to the task and carries no quantity (§4.3). What gets over-claimed is not the thing that looks numerical but the thing that looks like a reasoning step.

**Self-correction and repair.** The limits of intrinsic self-correction are well documented: Huang et al. (2023) find that without external feedback models struggle to identify and correct their own errors, and that performance sometimes degrades. §4.5's ladder is that finding with the content of the feedback varied. Tools and no hint helped unevenly — four of four models the first time and two of three the second, three of those gains small. Being told that a note was false repaired the answer for every model that cleared the instrumentation bar, both times; being told to put the context right, with the same handles it had already held for two rungs, then cleared the registered twenty-point ceiling in one measurement of seven and left the rest between -20 and +13 points, at  $n = 15$  each. The nearest published work is MemSecBench (Chen et al., 2026b), which measures whether an agent, told only to audit itself, removes poisoned long-term memory and keeps the benign kind. It measured that first, in the security framing, and two things differ. Its repair stage is scored on the state the memory ends up in, and what the poison went on to do is scored separately, at an earlier stage; this ladder scores the next answer to the task. And its agent is given no hint about which item is at fault and is never asked to name one, where here the subject names the item on the record before it is *asked* to change anything — it could already, having held the handles since unprompted — which is what makes naming and fixing separable at all, and lets the answer after being told and naming be compared with the answer after being asked to put the context right.

**Introspection and causal bypassing.** Binder et al. (2024) find that a model finetuned to predict its own behaviour beats a differently-trained model at it, and read that as privileged access; Lindsey (2026) injects concept vectors and finds that models can notice, the strongest positive result in the field. Binder et al. work on GPT-4, GPT-4o and Llama-3 and Lindsey on Claude models, none of which is in this cohort, and both required finetuning or white-box access where these measurements are zero-shot and black-box. Singh et al. (2026) is a recent reality check on that literature and is worth reading beside them. §4.4 asks a different question — not *what will I say* but *what is my answer made of* — and finds no own-context advantage over the same model reasoning about a matched dossier held by a second session that really ran. The two are compatible only if self-knowledge is not one faculty.

That null is what causal bypassing predicts (Morris and Plunkett, 2025): a report can be accurate by a route that never passes through the state it reports, and a subject here can get an ablation right by reasoning about the task — the Omsk annex is the decisive record, so removing it must change the answer — with no self-access involved. Predicts is not shows. What §4.4 reports is the absence of an advantage and not its sign, at five items per arm per model, with the foreign context arriving quoted where the subject's own arrives as context (§6). These numbers do not require privileged access to explain them; at this size they also do not rule it out, and they do not establish what the model is doing instead.

*This section was assembled by a language model. Three of its characterisations have since been checked by the works' own authors: Zeng et al. (2026), whose first author pointed me to the multi-turn track described above; VISTA, whose first author asked for no change; and MemSecBench, whose corresponding author asked that the scope of its repair stage be stated more exactly, which it now is. The rest has not been checked by anyone who*

*reads this literature regularly. It is the one part of the paper I cannot verify from my own data, and I would welcome corrections.*

*What I can now say is narrower, and worth stating exactly. Every identifier in §10 was resolved against arXiv on 2026-09-06 and matches the title and authors recorded there, so the works exist and their abstracts say what this section says they say. That is not the same as having read them closely enough to have placed them correctly, which is what the paragraph above is about. The check was prompted by a review in which a model with no network access declared every citation dated 2025 or later to be fabricated, on the grounds that the dates were in the future, the 2025 ones included. It ran `date`, was told the year was 2026, and did not revise, reading the agreement between the system clock, the commit history and the cohort's model names as evidence of a constructed scenario rather than of the date. All of it resolves. The episode belongs to §7 rather than to this section, except in one respect: it measured, and did not update.*

## 9. Reproducing this

This study is its own repository and holds the saved runs, the analysis and the write-ups. The runtime and the instrument are a separate one, `nachalnik`, which this depends on at a pinned commit (`d3b3ba6`) — deliberately, so that a second study starts from a harness rather than from a copy of this one, so that the instrument cannot learn what this study registered, and so that the command below reads the saved reports with the same code wherever it is run.

```
cargo run --release
```

reads `eval-runs/`, groups the reports by the instrument version that produced them, recomputes the tables in §4.3 for both collections, and reports each of the four replication thresholds against the value registered before the second collection — as a verdict, not a figure, because what counts as a replication was fixed in advance and printing only the number would hand that reading back to whoever is looking. It then reads the repair ladder over only the cells the planted falsehood fooled, which is the second reading in §4.5.

Nothing is re-requested and nothing costs anything. The collections themselves cost \$9.11 and \$15.09 — a similar number of requests on the experiments the two share, plus the harness check in §4.6, which the second ran on three models where the first ran it on one. `PREREGISTRATION.md` holds what was fixed in advance, with its §11 the dated log of every deviation; `RESULTS.md` and `RESULTS-v5.md` map each registered prediction to its outcome. Instrument versions 4 and 5; every experiment's exact wording is hashed into a digest printed with every result, so two numbers can be compared only when they were produced by the same questions. When the location probe came out, two of the seven digests moved and five did not.

`METHODOLOGY.md` here is the full methods document — every metric's definition, the controls, and the eleven known threats to validity, three of them uncontrolled. `RUNBOOK.md` is what was run, in what order, and what it cost.

## 10. References

Every arXiv identifier below was resolved against arXiv on 2026-09-06; the titles and authors are as recorded there. `RELATED.md` holds the longer positioning notes, including work this paper does not cite, and was checked on 2026-09-03.

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- 

## Appendix A: the arithmetic, by hand

The main claims are counts, and can be checked without any statistics at all.

**Red herrings.** Twelve per dossier-set per model, minus items whose ablation moved the answer for that model.

- first collection:  $12 + 12 + 10 + 12 + 11 + 12 = 69$ . Claimed: 0.
- second collection:  $12 + 12 + 10 + 11 + 10 + 11 = 66$ . Claimed: 0.
- both: 135. Claimed: 0.

**Plain inert notes.**

- first collection:  $16 + 18 + 16 + 17 + 16 + 18 = 101$ . Claimed: 0.
- second collection:  $18 + 16 + 16 + 17 + 16 + 15 = 98$ . Claimed: 1, by `deepseek-v4-flash`.
- both: 199. Claimed: 1.

**Numeric over-claims.** 32 in the first collection and 31 in the second, 63 in all. Every one of them fell outside the red-herring set and none on a red herring, which is the whole of the discrimination result — 71 such notes were measured the first time and 70 the second. Of the 63, **62 are notes belonging to the question's own arithmetic and one is not:** `mill/records/guidance` carries no quantity and is filed here only because its text contains a year (§4.3).

Under the preregistration's §8.2 gate, which §2.4 records as registered and not applied, the same counts are 26 and 25, 51 in all, with the same single exception among them. The red-herring count is 0 of 59 and 0 of 58; the plain count is 0 of 87 and 0 of 84.

**Location.** 18 questions per model, 6 models = 108. Correct: 2 (both from `solar-pro4`). Kept for completeness; §4.2 explains why this count does not measure what it looks like it measures.

**Sign test on direction.** Six models, all six with the same sign, under a null of "either direction equally likely":  $(1/2)^6 = 1/64 = 0.0156$ . That is the one-sided figure, and the right one for the second collection, where the direction was registered in advance. For the first, where it was not, the two-sided figure is  $2/64 = 0.031$ .

**Confidence intervals** in the repository's output are Wilson score intervals. The accuracy scores are widened for clustering by dossier where more than one dossier contributes, with the unadjusted interval printed alongside so the cost of the adjustment is visible. **The primary endpoint's intervals are not:** `Surface` is computed from four counts and is handed no cluster information at all, so its numeric and plain intervals are plain Wilson intervals over items that are in fact clustered by dossier, and are correspondingly too narrow. No claim in §4.3 or §5.1 depends on one — they are counts — but the intervals printed there should not be read as adjusted.